Paper Revisions: One Molecule, Two Markets

Every replacement for `Doc/Paper_draft.docx`, in paper order, with the exact text to cut, the text to put in its place, and where it goes. Prepared 25 October 2026 for Adhiraj Bhagawati and co-authors.

How to use this file. Each change gives a **Location** (section, paragraph and the words it starts with), a red **CUT** box with the current text, and a green **REPLACE WITH** or **INSERT** box with the new text. Text in green boxes is ready to paste. Citations in green boxes are in the paper's APA 7 style; every new reference is listed in section 10. Spelling follows the draft (US English).

Ground rule followed. Every number in the new text was checked against the source it cites during this revision, on or before 25 October 2026. Where something could not be checked, it is said so in the text or listed in section 11, never stated as fact.

1. At a glance

Nine requests from the economist and one removal, each mapped to the paper section that changes and to what already changed in the artifact (A = website, B = desktop game). The code changes are done in both conditions and both repositories.

#	Economist's request	Where it lands in the paper	What changed in the artifact
1	Argue against itself; demolish counterfactuals such as global price harmonization	1 (new paragraph 3); 2.6 (rewritten); 3.1 items 2 and 3	New module 2, "The Pipeline: Why Patents Exist" (A section 2; B `interlude-pipeline.js`). Court cases show what happened and both sides; no "Correct"/"Incorrect"
2	Positive case for patents before evergreening	2.6 paragraph order; 3.1 item 2 before the legal module	Pipeline sits before the courtroom; case order Gilead, Novartis, Bayer, Roche, Pfizer, Turing
3	Define price-comparison parameters; use recent data	1 paragraph 1; 3.1 item 1; 3.2.3 module 1 note	One day of imatinib 400 mg from one study: US $295.34 vs India $2.20 (134 to 1). 2024 Medicare Part D and 2026 Indian retail prices shown beside it
4	Global context: Africa, Southeast Asia; benchmark China, Brazil, South Africa	2.6 new paragraph; 3.1 items 6 and 10	A section 10 "Beyond the OECD" panel; B module 10 briefing; globe shows sourced export shares by region
5	Procurement and financing: US insurance and rebates vs India out-of-pocket	2.6 new paragraph; 3.1 item 4	A section 4 "Who pays, and how"; B record room card "WHO PAYS"
6	QALY instead of "value of a year"; coverage where marginal benefit equals marginal cost	2.6 QALY paragraph; 3.1 item 5; Table 1 row 5	A section 5 "QALYs and the Coverage Decision"; B ward coverage-rule wall; tooltip errors fixed
7	"S-shaped diffusion curve", not "epidemic model"	3.1 item 6	"Epidemic" and "contagion" removed; curve labelled illustrative
8	Sensitivity analysis of the game matrix	3.1 item 9; Table 1 row 9; new Table 3	Race has five exclusivity stages and a Stay-out option; payoffs and equilibria computed
9	Short blog or explainer	Not in the paper	`Doc/Explainer_Is_a_2_dollar_generic_too_cheap.docx`
R	Remove the supply-and-	3 intro; Fig. 1; 3.1 item 4;	Section and scene removed; the ward

#	Economist's request	Where it lands in the paper	What changed in the artifact
	demand chart and story	Table 1 row 4; 3.2.2; 3.2.4; Fig. 4	inherits its price from the market floor

2. Title, abstract and keywords

The title stays. One abstract sentence changes so the opening is precise and two-sided; the keywords gain two terms.

Location: Abstract, sentence 1.

CUT

> The difference in price for the same drug between countries, as little as pennies versus hundreds of dollars, comes down to a number of economic factors: patent protection, generics entering markets, compulsory licensing (where a government can authorize use of a patented invention without the patent holder's consent), and the balance between encouraging innovation and expanding access.

REPLACE WITH

> The same drug can cost a hundred times more in one country than in another: in one study, a day of the leukemia drug imatinib cost about $295 in the United States and about $2 in India. The gap comes from patent protection, the entry of generics, compulsory licensing (where a government authorizes use of a patented invention without the patent holder's consent), and a real trade-off between paying for the research that produces new medicines and widening access to the ones that exist.

Abstract sentence 2 ("Experts understand these concepts, but they stay invisible to most people.") stays.

Location: Keywords line.

REPLACE WITH

> explorable explanations; narrative visualization; serious games; gamification; mental models; health economics communication; pharmaceutical patents; cost-effectiveness

3. Section 1, Introduction

Four edits: tighten the Gleevec figures, state the comparison basis, add a counter-case paragraph, and drop supply and demand from the list of concepts.

Location: Paragraph 1, sentence 1 ("In 2014, one pill of Gleevec...").

CUT

> In 2014, one pill of Gleevec (imatinib), which is a leukemia drug from Novartis, cost $179.93 in accordance with US Medicare and Medicaid data, and this cost was up from $92.74 in 2010, as Novartis had raised the price several times since the drug's launch (Kodjak, 2016).

REPLACE WITH

> In 2014 one pill of Gleevec (imatinib), Novartis's leukemia drug, cost $179.93 in the United States, up from $92.74 in 2010, according to Centers for Medicare & Medicaid Services data reported by Kodjak (2016); the report does not give the pill's strength.

Location: Paragraph 1, immediately after the sentence ending "...as much as around US$5,000 per month (Gabble & Kohler, 2014)."

INSERT

> Measured on a single basis, imatinib 400 mg once a day priced per day of treatment, Hill et al. (2016) report $107,799 per person-year in the United States ($295.34 a day) and $803 per person-year for the lowest available Indian generic ($2.20 a day), so one US day of treatment costs as much as about 134 days in India.

Location: Paragraph 1, last sentence.

CUT

> Hence, in conclusion, the chemical makeup of the drug is the same everywhere; what changes is who controls the market and what the laws allow in each country.

REPLACE WITH

> The molecule is the same everywhere; what differs is who controls the market and what each country's law allows.

Location: New paragraph 3: after the Daraprim paragraph (ending "...(Hemphill & Sampat, 2012; Kyle et al., 2026; Sampat & Shadlen, 2018).") and before "Essentially, economists have studied...".

INSERT

> A low generic price is only half of the story, because a generic is cheap partly because someone else paid to invent the drug. Only 3.4% of oncology development programs in clinical trials between 2000 and 2015 reached approval (Wong et al., 2019), and the median capitalized research and development investment per approved antineoplastic or immunomodulating agent was $2,771.6 million in 2018 dollars, counting failed programs and the cost of capital (Wouters et al., 2020). A patent is how the inventor recovers that money, and without high prices somewhere fewer drugs would be developed. One global price would not solve the problem either. When low prices spread across borders through reference pricing and parallel trade, firms delay or skip launches in low-price markets (Danzon et al., 2005), whereas differential pricing, with high-income markets paying most of the shared research cost, can fund research and widen access at once (Danzon & Towse, 2003). The evidence also limits the argument for patents: their introduction in developing countries was not followed by more investment in diseases that mainly affect the poorest countries (Kyle & McGahan, 2012). The artifact therefore presents both sides, and the gamified version scores them against each other.

Location: Paragraph beginning "Essentially, economists have studied...", sentence 2.

CUT

> Supply and demand curves, the Lerner index, and an inverted-U relationship between market competition and innovation (Aghion et al., 2005) are each component parts

REPLACE WITH

> The cost of research and development, cost-effectiveness thresholds, the Lerner index, and an inverted-U relationship between market competition and innovation (Aghion et al., 2005) are each component parts

The rest of that sentence, the other Introduction paragraphs, RQ1 to RQ3 and the contributions stay as they are.

4. Section 2.6, Domain background

Location: All three current paragraphs of 2.6, from "Two economic studies supply the core..." to "...turn it into interaction."

CUT

[All three current paragraphs of 2.6.]

REPLACE WITH

The artifact starts where the economics of medicines starts: with the cost of inventing one. Only 3.4% of oncology development programs in clinical trials between 2000 and 2015 reached approval (Wong et al., 2019). Counting those failures and the cost of capital, the median research and development investment per new drug, in a sample of 63 drugs approved in the United States between 2009 and 2018, was $1,141.7 million, and $2,771.6 million for antineoplastic and immunomodulating agents, in 2018 dollars (Wouters et al., 2020); an estimate built from industry data puts the figure at $2,558 million per approved compound in 2013 dollars (DiMasi et al., 2016). The TRIPS agreement sets a patent term of at least 20 years from the filing date (World Trade Organization, 1994, Art. 33). A patent exists so that an inventor can recover such costs before rivals copy the product, and the artifact makes that case first, in a pipeline module placed before the legal one.

The same patent system can be stretched. Firms file secondary patents to extend exclusivity (Sampat & Shadlen, 2017), although challenges from generic companies have kept such patents from lengthening effective market life (Hemphill & Sampat, 2012). The legal module draws on work on how Section 3(d) reshaped Indian patent prosecution (Basheer & Reddy, 2008; Mueller, 2007; Sampat & Shadlen, 2018) and on what TRIPS-era product patents did to prices and access in India (Duggan et al., 2016; Kyle et al., 2026). The evidence on whether patents raise research in poorer countries is weaker than the case for patents in general: patent protection was associated with more research and development in high-income countries, but its introduction in developing countries was not followed by more investment in diseases that mainly affect the poorest countries (Kyle & McGahan, 2012).

The pricing mechanics rest on measured evidence. Generic prices fall sharply in the years after the first generic enters (Grabowski & Vernon, 1992) and keep falling as more generics join (Frank & Salkever, 1997). Where brand patents were challenged, average prices fell by about 20% after generic entry (Branstetter et al., 2016), and original manufacturers have responded with branded generics of their own (Reiffen & Ward, 2007). Chaudhuri et al. (2006) used Indian market data to estimate the welfare cost of patent protection for quinolone antibiotics, and Aghion et al. (2005) showed an inverted-U relationship between product market competition and innovation. The patent race module rests on the economics of duplicated research, in which competition can push research effort beyond the socially optimal level (Dasgupta & Stiglitz, 1980); we add a sensitivity analysis showing that the same race stops, and then stops producing a drug, as the prize shrinks.

A common counter-proposal is one global price. The literature points the other way. Because research is a cost shared across all markets, Ramsey-style differential pricing, with high-income markets paying markups that cover most of the shared cost and poorer markets paying less, is the second-best efficient way to fund it (Danzon & Towse, 2003). When low prices leak across borders through external reference pricing and parallel trade, firms delay or skip launches in low-price markets: across 25 major markets in the 1990s, only about 55% of potential launches of 85 new chemical entities took place (Danzon et al., 2005). Uniform pricing would tend to settle near the high-income price and leave low-income patients without the drug.

How a medicine is paid for changes what its price means. In India, out-of-pocket spending was 39.4% of total health expenditure in 2021–22 and government spending 48% (Ministry of Health and Family Welfare, 2024), so many patients pay the pharmacy price themselves. In the United States, list prices are reduced by confidential rebates and discounts; these gross-to-net reductions on brand-name drugs were estimated at $335 billion in 2023 (Drug

Channels Institute, 2024), and Medicare's first negotiated prices for ten drugs, 38% to 79% below list prices, take effect on 1 January 2026 (Centers for Medicare & Medicaid Services, 2024). A US list price therefore overstates what insurers pay, while an Indian pharmacy price is close to what a patient pays.

India is the largest generic supplier but not a special case. By the Department of Pharmaceuticals' account, India provides about 20% of global generic supply, nearly 40% of generic demand in the United States and over 70% of the global supply of antiretroviral drugs (Department of Pharmaceuticals, n.d.). Other middle-income countries have used procurement or licensing to the same end. China's "4+7" volume-based procurement pilot in 11 cities cut the winning bids for 25 drugs by an average of 52% against the 2017 minimum purchase price (Tang et al., 2019). Brazil issued a compulsory license on the HIV drug efavirenz on 4 May 2007 after rejecting Merck's offer of US$1.56 a day, when an Indian generic cost US$0.45 (Alcorn, 2007). Thailand issued compulsory licenses on efavirenz in 2006 and on lopinavir/ritonavir and clopidogrel in 2007 (Médecins Sans Frontières, 2006), and in April 2001, 38 pharmaceutical companies withdrew their court challenge to South Africa's 1997 Medicines and Related Substances Control Amendment Act (South African Government, 2001). The artifact keeps India as its focus and reports these as brief comparisons.

For the health economics module we use the field's standard tools (Drummond et al., 2015). A quality-adjusted life year (QALY) counts a year in full health as 1 and a year in poorer health as a fraction of 1. A payer compares treatments by the incremental cost-effectiveness ratio (ICER), the extra cost per extra QALY, and funds a treatment when that ratio is at or below a threshold. With a fixed budget, the threshold represents the health the money would buy elsewhere, the cost per QALY of the activity it displaces (Claxton et al., 2015; McCabe et al., 2008); spending is best allocated when the last QALY bought costs exactly that, where marginal benefit equals marginal cost. For imatinib against interferon-alpha plus low-dose cytarabine, a US model estimated 3.89 QALYs gained at $43,300 per QALY (Reed et al., 2004). India has no official threshold; a 2026 survey estimated willingness to pay at US$2,535 per QALY, 1 to 1.52 times GDP per capita (Chugh et al., 2026), and WHO authors caution against using multiples of GDP per capita as decision rules (Bertram et al., 2016). The US debate centers on a $50,000 benchmark (Neumann et al., 2014), and the wider question is whether medical innovation is worth what it costs (Cutler & McClellan, 2001). We use all of this as material for the artifact to teach and do not claim to extend it.

Order now runs: the case for patents; their limits and evergreening; pricing mechanics; the harmonization counterfactual; financing; global benchmarks; QALYs. Existing citations are kept.

5. Section 3 intro and 3.1, Condition A

The numbered list of ten sections is replaced in full: a new item 2 (the pipeline) goes in, the old item 4 (supply and demand) comes out, and the other items are renumbered and, where noted, rewritten. A new Table 3 follows the list.

Location: Section 3, paragraph 1, last sentence.

CUT

The supply and demand section, shown in both versions in Fig. 1, makes the difference concrete.

REPLACE WITH

The patent race section, shown in both versions in Fig. 1, makes the difference concrete.

Location: 3.1, the numbered list, items 1 to 10 (from "1. The molecular scrollyteller..." to the end of "10. Global benchmarks...").

CUT

> [Current items 1 to 10, including item 4, "Microeconomics: supply and demand".]

REPLACE WITH

> **1. The molecular scrollyteller.** A rotating 3D model of imatinib (Gleevec, PubChem CID 5291) switches between two markets. Both prices are for imatinib 400 mg once daily, priced per day of treatment from one study: $295.34 a day in the United States ($107,799 per person-year) and $2.20 a day for the lowest available Indian generic ($803 per person-year), so one US day costs as much as about 134 Indian days (Hill et al., 2016). The model turns red at the US price and green at the Indian one. The text beside it explains that the US price of a pill rose from $92.74 in 2010 to $179.93 in 2014 (Kodjak, 2016), that a US generic launched in 2016, and that the Supreme Court of India's 2013 ruling under Section 3(d) kept generics available (Gabble & Kohler, 2014; on Section 3(d), see Basheer & Reddy, 2008). More recent figures are shown beside the comparison but not mixed into it: in Medicare Part D in 2024, average spending per dosage unit was $233.96 for Gleevec and $33.39 for generic imatinib, before rebates (Centers for Medicare & Medicaid Services, n.d.), and on 25 October 2026 an Indian online pharmacy listed 400 mg imatinib tablets from Rs 25 to Rs 213.2 each (1mg, 2026).
>
> **2. The pipeline: why patents exist.** Two sliders set how many candidate cancer drugs enter clinical trials and how many years the winner keeps the market before generics enter. At a 3.4% success rate (Wong et al., 2019) the section shows the expected number of approvals. At a median capitalized cost of $2,771.6 million per approved antineoplastic agent (Wouters et al., 2020) it shows the net revenue the drug must earn in each year of exclusivity to repay its development, R = C r / (1 − (1 + r)^−T) at the 10.5% cost of capital the estimate uses: $337 million a year over 20 years, $741 million over 5. Beside it the section sets out the counter-case and its limits (Danzon & Towse, 2003; Danzon et al., 2005; Kyle & McGahan, 2012).
>
> **3. The legal labyrinth.** The reader sits as an Indian patent authority and rules on six real cases, in this order: Gilead's voluntary licensing of its hepatitis C cure (Gilead Sciences, 2014); Novartis and evergreening under Section 3(d) (Gabble & Kohler, 2014); Bayer against Natco over a compulsory license for Nexavar (Knowledge at Wharton, 2012); Roche against Cipla over an injunction on Tarceva (F. Hoffmann-La Roche Ltd. v. Cipla Ltd., 2008, 2009; Cipla Ltd. v. F. Hoffmann-La Roche Ltd., 2015); Pfizer's patent on Sutent, which was revoked (Balakrishnan, 2012; Vishwanathan, 2013); and, last, Turing's closed distribution scheme for Daraprim, the one case that involves no patent (Carrier et al., 2016; Gallant, 2015). No ruling is marked right or wrong. After each one the file shows what actually happened, the patent holder's case and the access case. The ruling on the imatinib case decides whether the market in the next section is open.
>
> **4. Macroeconomics and social returns.** Producer and consumer surplus shift as generic competitors enter. We model the price with a deliberately simple decay formula, P_N = P_0 × N^(−0.75), where N is the total number of sellers, including the original manufacturer. The formula comes close to the FDA's estimates for one and two generic competitors, which put prices about 39% and 54% below the brand, but it understates the larger falls seen when four or more generics enter, which can reach 79% or even more than 95% below the brand price (Conrad & Lutter, 2019). Whether the market is a monopoly is decided by the reader's ruling in the imatinib case. A financing panel contrasts India, where out-of-pocket spending was 39.4% of health expenditure in 2021–22 (Ministry of Health and Family Welfare, 2024), with the United States, where confidential rebates cut brand-name list prices by an estimated

$335 billion in 2023 (Drug Channels Institute, 2024) and Medicare's first negotiated prices take effect in 2026 (Centers for Medicare & Medicaid Services, 2024).

5. Health economics: QALYs and the coverage decision. A fixed $10,000,000 budget converts the price of treatment, which the reader sets between $30,000 and $2,000 a patient a year, into patients treated and QALYs lived. Each patient treated is credited with 7.34 QALYs over ten years, the figure a published cost-effectiveness model of CML reports for starting on imatinib (Yamamoto et al., 2019); the section shows the trial survival behind it, an estimated 83.3% overall survival at ten years (Hochhaus et al., 2017). Because these are QALYs lived rather than QALYs gained over a comparator, the section does not compute an ICER. It sets out the coverage rule instead, funding a treatment while its ICER is at or below a threshold that reflects the health the budget would buy elsewhere (Claxton et al., 2015), with two reference points: $43,300 per QALY for imatinib in the United States (Reed et al., 2004) and an Indian willingness to pay of US$2,535 per QALY (Chugh et al., 2026).

6. Technological diffusion and global trade. An animated map and an S-shaped (logistic) diffusion curve follow Indian pharmaceutical exports from about $3 billion in 2003 to about $27 billion in 2023 (Indian Pharmaceutical Alliance, 2025) and $30.47 billion in 2024–25 (Press Information Bureau, 2025). The timeline starts in 2003, the earliest year we could source, and the years between the reported points are interpolated. The curve's midpoint and slope are illustrative; they were chosen, not fitted to the export series. Clicking a destination shows its share of India's pharmaceutical exports by value in 2024–25: 37.6% to the NAFTA countries, 18.9% to Europe and 12.9% to Africa (Pharmexcil, 2025). A note on complementary assets records that low prices were not enough on their own: after the US FDA offered tentative approval for generic antiretrovirals bought with PEPFAR funds, generics rose from 14.8% of the antiretroviral packs PEPFAR procured in 2005 to 89.3% in 2008, and 9 of the 12 manufacturers with tentative approval were based in India (Holmes et al., 2010). An optional view shows India's imports of active ingredients from China, which do not measure India's overall dependence on China, because India produces much of its own supply (Wosińska & Shi, 2025).

7. The human metric: affordability and the Lerner index. A month of treatment is converted into the days of work it costs, in rupees and at the wages in force when the compulsory license was granted in March 2012: Rs 120 a day, the MGNREGA wage notified for Uttar Pradesh from 1 January 2011 (Ministry of Rural Development, 2011), or Rs 258 a day, the average daily earnings of a male regular wage or salaried employee in the urban informal sector in 2011–12 (National Sample Survey 68th round, as reported in Business Standard, 2014). At Rs 2.8 lakh a month for Bayer's Nexavar and Rs 8,800 for Natco's generic (Knowledge at Wharton, 2012), a month costs the rural laborer 2,333 days of work at the patented price and 73 at the generic price, and the urban worker 1,085 and 34. The result is compared with the WHO/HAI benchmark, under which a month's supply of a medicine for a chronic condition counts as affordable if it costs one day's wage or less (World Health Organization & Health Action International, 2008, p. 127). The same prices in dollars, $5,500 and $171 a month (Knowledge at Wharton, 2012), feed the Lerner index of monopoly power, $(P - MC)/P$. With a marginal cost of $74.18 a month, the full cost of production for sorafenib 400 mg twice daily before the 50% mark-up in Hill et al. (2016, Table 1), the index is 0.987 for the patented drug and 0.566 for the generic.

8. Antitrust and innovation: the inverted-U. The module pairs a market-concentration index (HHI) with an inverted-U curve relating innovation to competition (Aghion et al., 2005), drawn over the number of competing firms as a simplification of their Lerner index measure of competition.

> **9. Game theory: the patent race.** A payoff matrix for two firms that can invest High ($400 million), Low ($100 million) or stay out, racing for one winner-take-all patent. With a $1,000 million prize both firms overspend on duplicate research, since competition in R&D can push research effort beyond the socially optimal level (Dasgupta & Stiglitz, 1980). A sensitivity control shortens the time before generics enter, which shrinks the prize, and the equilibrium moves through five stages (Table 3).
>
> **10. Global benchmarks: the generic paradox.** OECD figures show that generics can make up most of a market by volume while taking a much smaller share of its value. In Germany's reimbursed market in 2021, generics were 83% of volume but only 29% of value, against OECD averages of 54% and 25% (OECD, 2023, Figure 9.7). The chart shows five systems (Chile, Germany, the United Kingdom, the OECD average and Switzerland), using the data labels printed in that figure; market coverage differs by country, as the figure notes state. The United States and Japan were removed because their values had no source. A second panel gives brief comparisons outside the OECD: China, Brazil, Thailand and South Africa (Tang et al., 2019; Alcorn, 2007; Médecins Sans Frontières, 2006; South African Government, 2001).

Item 8 is the draft's current text, unchanged. Items 1, 4, 6, 7 and 10 differ from the earlier revision map: they now carry the data verified on 25 October 2026 (recent prices, the CMS announcement, sourced regional shares, the PEPFAR evidence, rupee wages, and the OECD figure labels).

Location: New Table 3, placed after the 3.1 list.

Caption: Table 3. Sensitivity of the patent race to the time before generic entry. The prize is V(T) = 1000 × (1 − 1.105^−T) / (1 − 1.105^−20) $ million: the same yearly profit over fewer years, discounted at 10.5%. Strategies: High ($400M), Low ($100M), Out ($0). Payoffs: V − C for the higher or only investor, V/2 − C for equal investors, −C for the lower investor, 0 for Out.

Years before generic entry (T)	Prize V ($M)	Pure-strategy equilibria	Payoff in equilibrium ($M)	Reading
20	1,000	(High, High)	100 each	Wasteful race: both would earn 400 each at (Low, Low)
10	731	None	—	No stable pair of choices
5	455	(Low, Low)	127 each	The race stops
1	110	(Low, Out), (Out, Low)	10 to the investor, 0 to the other	Room for only one firm
0.5	56	(Out, Out)	0	Nobody invests; the drug is not developed

Sentence under Table 3: "High beats Low against either rival choice only when V exceeds twice the cost gap, 2 × (400 − 100) = $600 million; below $800 million the race no longer pays for two High programs, and below $100 million it pays for none."

Each row was recomputed from the formula (V(10) = 730.8, V(5) = 454.7, V(1) = 109.9, V(0.5) = 56.3, rounded in the table) and matches the artifact's output.

6. Section 3.2, Condition B

Five places change: the objectives sentence, the scoring sentence, Table 1, the module notes, and the consequence chains. Sections 3.2.1, 3.2.5 and 3.2.6 stay as written (there are still four record rooms and ten modules).

Location: 3.2, opening paragraph, after the sentence "Because nothing is computed in a second place, the two conditions cannot drift apart on the numbers."

INSERT

> Every scene and record room reads its figures from one shared model object in the page script, so the explorable and the game show the same numbers by construction.

Location: 3.2.2, Explicit objectives, sentence 2, the end of the sentence.

CUT

> treating at least 2,000 patients on the fixed budget, bringing the clearing price down to $70 or below, or parking the market between three and eight firms.

REPLACE WITH

> treating at least 2,000 patients on the fixed budget, funding enough candidate drugs to expect one approval, or parking the market between three and eight firms.

Location: 3.2.2, Scoring, sentence 2, the end of the sentence.

CUT

> ruling for a monopoly raises Innovation and lowers Access, and opening a market does the reverse.

REPLACE WITH

> upholding a patent holder's claim raises Innovation and lowers Access, intervening does the reverse, and funding the pipeline raises Innovation.

Location: Table 1: replace the whole table body. Keep the caption.

Module	Economic concept	In Condition A	In Condition B: gesture and objective
1	Price versus cost	Toggle the US and Indian price on the molecule	Drag one sample onto the other, then a banknote into a dispenser: prove the samples are one compound, and see that one US day of treatment buys 134 days in India
2	R&D attrition, cost and exclusivity	Move a candidates slider and an exclusivity slider; read expected approvals and the yearly revenue needed	Load candidates into a trial rack and run the trials, then turn an exclusivity dial: fund enough candidates to expect one approval
3	Patent law and rulings	Read each of six cases and its outcome, with both sides' case	Pick up the case file, then bring a gavel down on a ruling paper: rule on all six cases
4	Producer and consumer surplus	Drag a competitor count upward	Haul a gate lever along its rail and turn a procurement wheel: make social access exceed private profit
5	Cost-effectiveness, QALYs and the coverage rule	Move a cost slider and read patients treated and QALYs lived	Turn a price wheel under a wall that states the coverage rule: treat at least 2,000 patients on the fixed budget
6	Diffusion and trade	Scrub a 2003–2024 timeline on a flat map	Drag a timeline and click destinations on a globe: reach 2012 or later and inspect a market
7	Affordability and market power	Read days of labor for a given price	Flip two switches: bring the price to 90 days of labor or fewer
8	Competition and innovation	Read the inverted-U curve	Add and remove firms to roll a ball along the curve: park the market between three and eight firms
9	The patent race and its sensitivity	Choose a price-decay stage, then High, Low or Stay out,	Commit a stack of chips (or the empty tray) as a sealed bid, turn the exclusivity dial, and

Module	Economic concept	In Condition A	In Condition B: gesture and objective
		and read the 3 × 3 payoff matrix	commit again: play at two prize sizes
10	The generic paradox and comparable economies	Read an OECD comparison chart and four non-OECD comparisons	Read the four comparisons in the briefing, inspect pedestals and lock in an answer: identify the system with the widest volume-to-value gap

Location: 3.2.3, module 1 note, sentence 3.

CUT

> Feeding it into the dispenser fills one tube with a single pill and the other with about 82 pills.

REPLACE WITH

> Feeding it into the dispenser fills one tube with a single day's 400 mg dose and the other with 134, the ratio of the US and Indian prices per day of treatment in Hill et al. (2016).

Location: 3.2.3, the module 2 note (the courtroom) becomes the module 3 note. Change "In module 2, each case arrives..." to "In module 3, each case arrives..." and replace the note's last sentence.

REPLACE WITH

> Once the ruling is made, the same document shows what actually happened, the patent holder's case and the access case; no ruling is marked right or wrong. We kept all of the case material inside the scene as objects, with no panel laid over it.

The module 8 note is unchanged.

Location: 3.2.4, Choice and consequence, paragraphs 2 and 3 (from "The first chain runs from the courtroom..." to "...and no way to bring it down.").

CUT

> [Paragraphs 2 and 3 of 3.2.4.]

REPLACE WITH

> The first chain runs from the courtroom to the market floor. The market modeled downstream is imatinib's, so the ruling in the imatinib case (Novartis v. Union of India) sets its status: if the player grants the patent, the gate on the market floor is bolted and no competitor can enter. In the second, the price index at which the player leaves the market floor is the share of the patented price the health budget has to pay, so the ward's price wheel starts at that share of $30,000. In the third, the time before generic entry that the player chose in the pipeline sets the size of the prize when the patent race opens.
>
> The fourth chain reaches furthest, from module 3 to module 7. The cheap generic in module 7 exists only because a compulsory license was granted in the Bayer case, so a player who refused that license reaches the wage floor to find the switch to the cheaper market welded shut, the tower of working days standing where it is, and no way to bring it down.

Paragraph 4 ("Where a chain makes an objective unreachable...") stays.

7. Figures

Figure	Action	New caption
1	Replace the supply-and-demand pair with the patent race in both conditions. File:	Figure 1. Side-by-side: the same content in both conditions. Left, Condition A's patent race: a price-

Figure	Action	New caption
	`Doc/figure1_patent_race_A_vs_B_DRAFT.png`. Draft only: the Condition B half was rendered by a software renderer. Recapture it from the built 1.6.0 app (module 9, after committing at 5 years) before submission.	decay stage, three choices and a 3 × 3 payoff matrix with the equilibrium outlined. Right, Condition B, where the same matrix stands behind a table and the player commits a stack of chips and turns an exclusivity dial.
2	Keep. The architecture still has ten modules.	Unchanged
3	Keep. None of its five panels shows the removed clearing house. Optional: swap one panel for the pipeline.	Unchanged, or add "the trial rack in the pipeline" if a panel is swapped
4	Replace. File: `Doc/figure4_consequence_graph.png` (vector: `figure4_consequence_graph.svg`).	Figure 4. The consequence graph: which modules feed which, and what is inherited. Four chains: the imatinib ruling sets whether the market floor is open; the market's price index sets the ward's starting cost; the pipeline's years before generic entry set the patent race's prize; the Bayer license ruling decides whether a generic exists on the wage floor.

8. Section 4, Methodology and limitations

Location: 4.3, Test of Comprehension, sentence 1, the topic list.

CUT

> price vs cost, generic entry price decay, value of cheaper treatment in life years, Lerner index, and the innovation–access trade-off.

REPLACE WITH

> price versus cost, the cost and attrition of drug development, generic entry price decay, QALYs and the coverage threshold, the Lerner index, and the innovation–access trade-off, including how the patent race changes as exclusivity shortens.

If the item bank already has a supply-and-demand item, drop it or rewrite it as a pipeline item, for example: "If only about 3 in 100 cancer drugs that enter trials are approved, what has to be true of the price of the ones that are?"

Location: 4.7, Pilot, after the last sentence ("...and it is where the time box for artifact use is set.").

INSERT

> The pilot also times the pipeline module, which was added after the time box was set. Time on task is read from the session log, which records the time each section of Condition A is on screen and the time spent in each scene of Condition B.

A step-by-step pilot timing protocol is in Appendix A of this file. The pilot itself still has to be run with people; it was not run in this revision.

Location: 4.8, Limitations, last sentence (from "Finally, two sets of figures in the artifact remain unsourced..." to the end of the paragraph).

CUT

> Finally, two sets of figures in the artifact remain unsourced: the volume and value shares for the five health systems other than Germany and the OECD average, and the export values for the years between 2003, 2023 and 2024–25, which are interpolated between the reported points.

REPLACE WITH

Finally, some figures are less firm than others. The export values between 2003, 2023 and 2024–25 are interpolated between reported points. The Indian imatinib price is read from a figure, not the text, of Hill et al. (2016), and both imatinib prices come from a single study published in 2016; the artifact shows 2024 Medicare figures and 2026 Indian retail listings beside them but does not mix them into the ratio. The OECD shares are the rounded data labels of the published figure, and the OECD database has since revised the United Kingdom's series. The wages in the affordability module are 2011–12 figures, chosen to match the 2012 prices, so they describe the year of the compulsory license rather than today. The $1,000 million, $400 million and $100 million in the patent race are illustrative, chosen to show a pattern rather than to estimate any real market. The health economics module states the coverage rule and two published reference points but does not compute an ICER for India; we found no Indian cost-effectiveness model of imatinib for chronic myeloid leukemia on which to base one.

9. Corrections to claims the code contradicted

Not on the economist's list, but a reviewer would find them. Each is fixed in the code, and the paper text above already reflects the fix.

Claim or figure	What the old build did	Fixed to	Source checked
3.2: "nothing is computed in a second place"	`game.js` and the fourth record room recomputed affordability with Nexavar at $5,000 and the generic at $105, and the Lerner index with MC = $20	One shared model object (`window.ModelData`) that every scene and record room reads	—
3.1 item 7: MC $80, Lerner 0.985 and 0.532	Used $80 a month	$74.18 a month; Lerner 0.987 and 0.566	Hill et al. (2016), Table 1
Nexavar price $5,000	Page label and Bayer claim said $5,000	$5,500 (Rs 2.8 lakh)	Knowledge at Wharton (2012)
3.1 item 7: wages $3.20 (MGNREGA, "about Rs 267") and $15	Neither figure was sourced, and 2012 prices were divided by wages of a later, unstated year	Rupees, 2011–12: Rs 120 (MGNREGA, Uttar Pradesh) and Rs 258 (urban informal-sector salaried). Days of labor 2,333 / 73 (rural) and 1,085 / 34 (urban)	MoRD notification S.O. 82(E); Lok Sabha Q. 3591 (2012); Business Standard (2014) on NSS 68th round
3.2.4: market follows the last ruling	Market status followed whichever ruling came last; re-reading a case reset it	Set only by the imatinib ruling	—
OECD values for Chile, UK, Switzerland, OECD average	82.3/67.2, 78.4/34.3, 22.1/14.1, 54.1/24.7 (volume/value); these do not match the published figure	90/65, 80/54, 23/19, 54/25; Germany 83/29	OECD (2023), Figure 9.7 data labels, checked 25 Oct 2026
Module 10 answer	Widest gap was the United States (91.0 vs 18.0), which had no source	USA and Japan removed; answer is Germany (83 vs 29, gap 54 points)	OECD (2023), Figure 9.7
Globe module regional claims	"20% of global generic exports by volume", "over 600 FDA-approved sites", "over 40% of the US generic market", "over 80% of antiretrovirals", shares 34% / 19% / 16%; all unsourced	Export shares FY2024–25: NAFTA 37.6%, Europe 18.9%, Africa 12.9%; "about 20% of global supply", "nearly 40% of US generic demand", "over 70% of antiretroviral supply"; the FDA-sites claim	Pharmexcil (2025) Table 2; Department of Pharmaceuticals Annual Report 2025-26, p. 3

Claim or figure	What the old build did	Fixed to	Source checked
		was dropped	
Diffusion note on PEPFAR	"NGOs (like PEPFAR) built the cold-chain distribution networks"; unsourced, and PEPFAR is a US government program	PEPFAR generic share of ARV packs 14.8% (2005) to 89.3% (2008); 9 of 12 tentatively approved makers in India	Holmes et al. (2010), JAMA
Module 1: "about 82 pills"	Compared a per-pill price of unstated strength with a per-day price	Per day, same dose, same study: $295.34 vs $2.20, ratio 134	Hill et al. (2016), Discussion and Figure 3A (checked against the figure image)
Court outcomes "Correct"/"Incorrect"	Every patent grant was marked incorrect	No verdict; each case shows what happened and both sides	Gilead (2014); Wharton (2012); Delhi HC (2009); Business Standard (2012); FTC
Gilead case figures	"11 manufacturers, 101 nations", "$84,000 vs $300–900"	"Seven Indian manufacturers, 91 developing countries"; prices dropped	Gilead press release, 15 Sep 2014
Turing case	Presented a US case as if decided in India, with an unsourced price	States the case was in the US and what the FTC and courts did; Indian law cited as Competition Act s. 4	FTC case page; Competition Commission of India
QALY tooltip	"patients × 5 added years", "years of life saved"	"patients × 7.34 QALYs", QALYs lived, not gained	Yamamoto et al. (2019)
OECD scene labels	"Share of prescriptions", "share of spending"	"Generic share, volume" and "value", as OECD defines them	OECD (2023), Definition and comparability box
"Gleevac", "Imanitib"	Misspelled 11 times	Gleevec, imatinib	—
Wouters et al. (2020) median	Widely quoted $985M predates a 2022 correction	$1,141.7M median; $2,771.6M for antineoplastic agents	JAMA correction, 2022;328(11):1111

10. References to add

APA 7, to be merged alphabetically into the existing list. No existing reference is removed. Bibliographic details for journal articles were checked against Crossref on 25 October 2026; every other entry was opened at the URL given.

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Two notes on dating. The Department of Pharmaceuticals report carries no publication date on the copy opened, hence (n.d.). Crossref lists Danzon, Wang & Wang as published online in 2004 in the 2005 volume; APA uses the volume year.

11. Decisions taken and what is still open

Resolved in this revision

Item	Resolution
Indian imatinib price, $803 per person-year	Verified against Hill et al. (2016), Figure 3A, panel A ("India (generic)" $803). Image saved as `Doc/evidence/Hill2016_BMJOpen_Fig3_annual_prices.jpg`.
Recent prices	CMS Part D 2024 (Gleevec $233.96, generic $33.39 per dosage unit, before rebates) and 1mg listings of 25 Oct 2026 added beside the comparison, not into the ratio. No NPPA ceiling price was found for the 400 mg tablet; the one order found covers the 100 mg tablet only.
OECD values	Replaced with the data labels of Figure 9.7. Image saved as `Doc/evidence/OECD_HaaG2023_Fig9-7_generics_share_2021.jpg`. Germany (83.4/29.2), Chile (89.4/64.5) and Switzerland (22.7/18.5) also match the live OECD database; the UK series there has since been revised (70.0/27.8).
Globe regional claims	Replaced with Pharmexcil (2025) and Department of Pharmaceuticals figures; "600 FDA sites" removed.
PEPFAR cold-chain claim	Cut; replaced with Holmes et al. (2010).
MGNREGA Rs 267 / $3.20 and $15	Replaced with 2011–12 rupee wages (Rs 120, Rs 258) to match the 2012 prices.
FDA price-decay figures	Rechecked: 39%, 54%, 79%, more than 95% (AMP basis), Conrad & Lutter (2019).
Medicare negotiation	Now cited to the CMS press release of 15 Aug 2024: "38 to 79 percent discounts off of list prices", effective 1 Jan 2026.
Reference details	Chugh 457–466; Claxton 1–504; Danzon & Towse DOI 10.1023/A:1025384819575; Danzon, Wang & Wang: Y. R. Wang; Kyle & McGahan 94(4), DOI 10.1162/REST_a_00214.
Indian ICER	Not computed. Two searches found no Indian cost-effectiveness model of imatinib for CML; the limitation says so.
Race dollar figures	Kept illustrative (recommended). Tying them to the $2,771.6M pipeline cost would make the two modules one model, but every threshold in Table 3 would move and the matrix would no longer read in round numbers; the limitations paragraph states they are illustrative. Confirm with the economist.
Blog / explainer	Written: `Doc/Explainer_Is_a_2_dollar_generic_too_cheap.docx`.
Explainer guide and devlog	`Website_Explainer_Guide.docx` rewritten for the ten current sections (old copy kept as `Doc/Website_Explainer_Guide_v1.4_old.docx`); `CONDITION-B-DEVLOG.md` has a 1.6.0 entry and a corrected module table.

Still open (needs a person, or could not be confirmed)

- **Pilot timing.** Must be run with two or three people (Appendix A). It was not run here.
- **Figure 1, Condition B half.** Recapture from the built app at full resolution.
- **MGNREGA rate continuity.** Rs 120 was notified from 1 January 2011 and a revision took effect on 1 April 2012. No revision between those dates was found, but none was ruled out either; confirm before submission.
- **Urban wage source.** Rs 258 is taken from Business Standard's report of the NSS 68th round (informal sector, urban males). The NSSO report table itself was not opened.
- **1mg prices** are a one-day snapshot of retail listings and change often.

- **Department of Pharmaceuticals figures** (20%, 40%, 70%) are the government's own statements, cited as such, not independent estimates.

Appendix A. Pilot timing protocol

Purpose: set the time box for artifact use (procedure step 4) now that Condition A has one more section and Condition B one more scene. Runs before the main study; pilot data are not pooled.

1. **Participants.** Two or three students from the target population, at least one per condition; if three, the third does both conditions in counterbalanced order on separate days.
2. **Build.** Condition A: `index.html` served from the repository. Condition B: the 1.6.0 desktop build. Start each with `?pid=PILOT01` (A) or the participant id typed at the opening briefing (B), so the session log is named.
3. **Instruction.** "Work through the material at your own pace until you are done. Think aloud as you go." No time limit is announced and none is enforced.
4. **Record.** Screen and audio, plus the session log (`il-log.js`): per section of Condition A, time on screen and number of entries; per scene of Condition B, time in the scene, including briefings and record rooms.
5. **Stop rule.** The participant says they are done, or reaches the closing scene (B) or the last section (A). Note any point where they stall for more than two minutes without acting.
6. **Measure.** From the log: total time on task; time per module; the pipeline's share of the total. Report median and range per condition.
7. **Decide.** If the slowest pilot run of Condition B finishes within 30 minutes, keep the 25–30 minute box. If not, take the paper's stated fallback: lift the box for both conditions, or let time on task vary and report it as a covariate.
8. **Revise.** Fix any instrument wording the pilot flags, re-time only the changed parts, and record the final box in section 4.6 before the main run.

Appendix B. Sources checked in this revision

Opened on 25 October 2026 unless stated, with what was checked.

Source	What was checked
Hill et al. (2016), BMJ Open, Figure 3A image	US $107,799 and India (generic) $803 per person-year, imatinib; sorafenib panel C
CMS, Medicare Part D Spending by Drug (API data set, years to 2024)	Gleevec $233.96 (2024), $249.77 (2023); generic imatinib $33.39 (2024), $49.99 (2023), per dosage unit, weighted
CMS press release, 15 Aug 2024	"38 to 79 percent discounts off of list prices"; effective 1 Jan 2026; ten Part D drugs
Conrad & Lutter (2019), FDA	39%, 54%, 79%, more than 95% (AMP basis)
1mg substitutes listing for Imat 400	Glivec 400 Rs 213.2; range Rs 25 to Rs 202.6; Imatib 400 Rs 91.5 per tablet
OECD, Health at a Glance 2023, "Generics and biosimilars", Figure 9.7 image and text	Data labels for 26 countries and OECD26; figure notes on market coverage
OECD SDMX API, HEALTH_PHMC@DF_GEN_MRKT	Germany, Chile, Switzerland, UK 2021 values

Source	What was checked
Pharmexcil Handbook (14 June 2025), Table 2	Regional export shares FY2024–25 and total US$30,466.85M
Department of Pharmaceuticals, Annual Report 2025-26, p. 3	20% of global generic supply; nearly 40% of US generic demand; over 50% of Africa's generics; over 70% of global ARV supply
Holmes et al. (2010), JAMA, abstract and comment	14.8% to 89.33% of packs; $323,343,256 saved; 9 of 12 makers in India
MoRD notification S.O. 82(E) (India Water Portal copy)	Uttar Pradesh Rs 120, effective 1 Jan 2011
Lok Sabha Unstarred Q. 3591, 26 Apr 2012	Revised rates notified 23 Mar 2012, effective 1 Apr 2012
Lok Sabha Unstarred Q. 456, 22 Jul 2025	State notified rates FY2020-21 to 2024-25 (used only to confirm no national average is published there)
Business Standard, 8 Sep 2014	Urban informal-sector male regular wage/salaried: Rs 258 a day, 2011-12
Crossref records	Volume, issue, pages and DOI for Chugh, Claxton, Danzon & Towse, Danzon et al., Kyle & McGahan, Wouters, DiMasi, Reed, Bertram, Hochhaus, Hill, Tang, McCabe et al., Dasgupta & Stiglitz, Aghion, Gabble & Kohler, Holmes, Wong